

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, adherent	Molecules Electroporated	DNA: plasmid, 12 kB
Species Used	Mouse, embryonic stem cells		

Before the Pulse

Cell Growth Medium	DMEM + L Glutamine + Pen/Strep + Fetal Calf Serum + mercaptoethanol + non-essential amino acids. (GIBCO/BRL, Sigma)	Growth Phase at Harvest	Exponential growth phase
		Pre-pulse Incubation	Not incubated pre-pulse
Wash Solution	Phosphate Buffered Saline		

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	Room temperature		
Electroporation Medium*	Phosphate Buffered Saline	Cuvette Gap	0.4 cm
Cell Density	1.5 to 2.0 x 10 ⁷ cells / pulse	Voltage	0.160 kV
Volume of Cells	0.5 ml	Field Strength	0.4 kV/cm
DNA Concentration	1 µg / ml	Capacitor	960 µF
DNA Resuspension Buffer	Phosphate Buffered Saline	Resistor	(Pulse Controller) Ω none
Volume of DNA	100 µg / pulse	Time Constant	14.5 msec
After the Pulse			
Outgrowth Medium	DMEM + L Glutamine + Pen/Strep + Fetal Calf Serum + mercaptoethanol + non-essential amino acids.		

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature	37 °C
Length of Incubation	2 weeks and more
Selection Method or Assay Used	G418
Electroporation Efficiency	Not given
Per Cent Survival	2.38 x 10 ⁻⁵

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